

ful attempts at standardisation. For instance, Robot Operating System (ROS), an open source framework, has a certain set of libraries and tools that are flexible for writing software for robots. Robots supporting ROS have significantly increased over the years. "As this field is so nascent, standards are still not up in place. There are several other architectures being followed, but there does not seem to be any universally agreed upon and followed standard in place," says Sinha. This often results in robots being built from scratch for different applications. Therefore there is a lot of scope for standardisation, and ROS is one of the leading examples.

From a design perspective too, standardisation is yet to take place. The reason for no standardisation is partly varying customer requirements. Sinha says, "In some cases, there might be a need for a very high degree of standardisation, like when building a system for space missions. We are jointly working with Indian Space Research Organisation (ISRO) on a module for Chandrayaan-2 mission for the moon-landing part."

Standardisation in hardware and software must go hand in hand. In the electronics industry, a development board is usually compatible and works on various standard operating systems (OSes) such as Linux or Windows. There is a requirement for boards to be compatible with standardised robotics OSes. Sinha says, "Currently most of the boards like Raspberry Pi 2 and Arduino are crafted for the development of electronics and Internet of Things (IoT) designs. We need boards designed purely to

## Point of view: The modular approach

"To address the variance in requirements, we keep a range of highly-configurable sub-systems such as types of joint modules, links, wheel modules, gripper modules and animatronic faces with common mechanical and communication interfaces.

"On software side, we maintain customisable application programming interface (API) bundles for kinematic and dynamic simulation and control, advanced perception (through sensor fusion), machine vision, motion planning (for both manipulation and localisation), natural language processing (NLP) and machine learning. With this approach, it is possible to provide quick solutions for any complex problem in robotics."

Jayakrishnan T., CEO,  
ASIMOV Robotics Pvt Ltd



make robots." He adds, "An ideal robotics board must support all the common sensors and motors, typically with plug-n-play attachments." With standardised boards and OSes for robots, creating these will become exponentially easier.

## Long way to go for robots to mimic humans, human-machine cooperation

Scientists and researchers have developed robots over the years that mimic humans. Emotional Robot with Intelligent Network (ERWIN) was created by scientists from the UK. This robot could display five distinct emotions.

Another group of researchers from Massachusetts Institute of Technology, USA, has developed a robot named HERMES that can mimic moves made by a person to whom the feedback-balance interface is connected. Robots are evolving.

Due to demographics and lack of interest in blue-collar jobs, there will be a huge resource deficit in the

personal and professional services sector in the near future, believes Jayakrishnan. He says, "A massive moment is silently taking place globally among the leading technology organisations to address this issue through robotics." He adds, "A collaborative work force is the next big thing where humans and robots co-exist and co-operate in all possible ways."

Better standardisation will encourage innovation and competition, thereby enabling the creation of more effective designs. Sinha says, "We have the necessary hardware to process huge amounts of data but we are unable to make robots think like human beings. The question I often ask is: How can we make robots think and learn like humans?" We are already seeing some light in the form of deep learning implemented over ten layers of artificial neural networks, but still have a long way to go before robots come closer to human learning capabilities, according to Sinha. ●

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